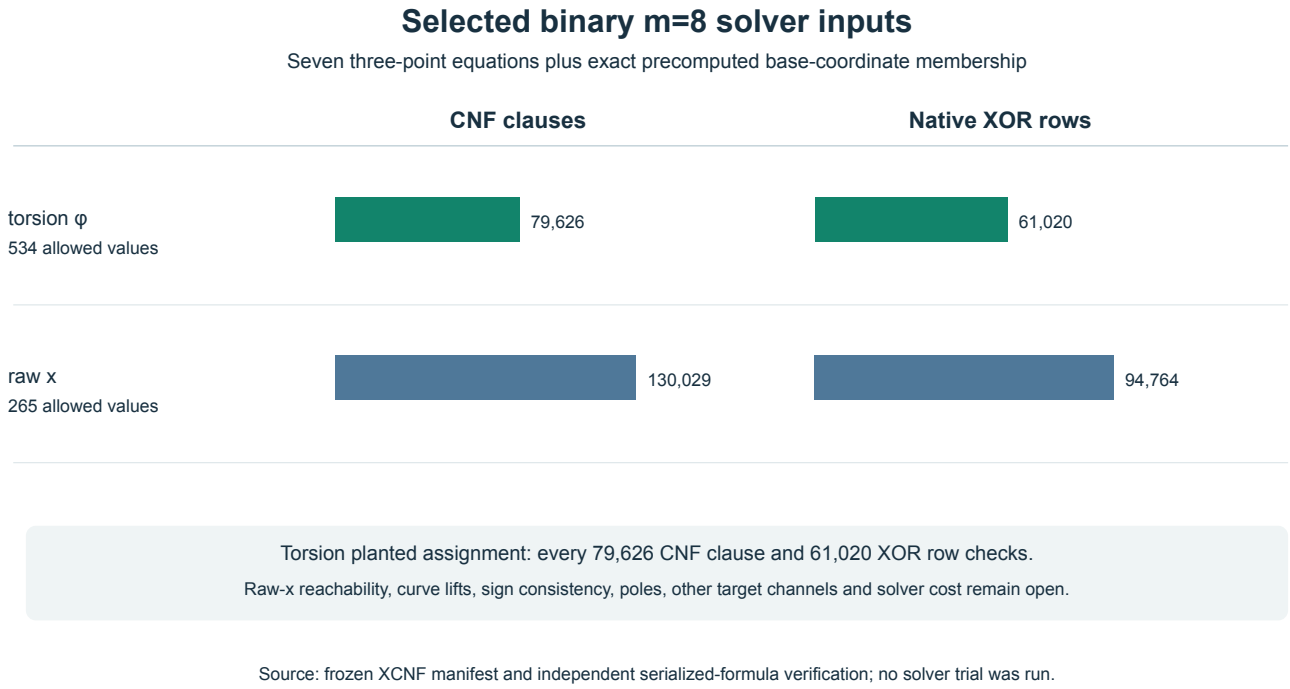
Membership-constrained binary m=8 circuit inputs

2026-10-17 · Prepared and checked solver inputs, without a solver trial

On the certified ordinary-binary GF(2^61) fixture, two selected bases feed seven three-point equations for one raw target channel. Every external coordinate is constrained to its exact target-blind accepted-value set. Live AND gates become three CNF clauses; XOR gates remain native parity rows; all 427 equation output bits are fixed to zero. This is a fixed schoolbook circuit encoding, with no curve-lift, sign, pole or other-channel branches.

Selected arm	Allowed / space	CNF variables	CNF clauses	XOR rows
Torsion φ	534 / 1,024	86,559	79,626	61,020
Raw x	265 / 512	137,744	130,029	94,764



Source: frozen XCNF manifest and independent serialized-formula verification; no solver trial was run.

A separate parser checked both compressed formula hashes, every gate definition, all eight complete coordinate-membership truth tables per arm, and the serialized formula's counts. For torsion φ, the archived planted primary assignment extends to every gate and satisfies all 79,626 CNF clauses and 61,020 XOR rows. The underlying point sum had independent Sage verification. The raw-x arm has no supplied satisfying assignment for that torsion-selected target; no satisfiability conclusion follows. Retained tiny format controls confirmed the installed CryptoMiniSat binary's `x` and `x-` parity syntax; neither selected formula was given to a solver.

Boundary. Additional point lifts, signs, poles, three other target channels, root certificates, useful-rank deduplication, linear algebra and complete phase costs remain. Exact source, gzip artifacts, digests and verification: `docs/reports/2026-10-17-binary-circuit-xcnf.md`.